

Week 3: Preparing the armed conflict data

Data preparation

The panel contains 186 countries for 2000–2019. The covariate table was used as its base. Four World Bank mortality tables were converted from wide to long format with one reusable function and joined by ISO code and year.

Disaster records were restricted to earthquakes and droughts in 2000–2019. Multiple events of the same type in a country-year were collapsed into one binary indicator.

Conflict event deaths were summed within country, year and conflict ID. A country-year was coded as exposed if at least one conflict reached 25 battle-related deaths. The conflict year was increased by one before merging, so the outcome in year t is matched to conflict in year $t-1$. This follows the UCDP conflict definition and one-year lag described in Jawad et al. (2021), Methods, pp. 4–6, doi:10.1371/journal.pmed.1003810.

There are 2,696 source rows with both conflict ID and death count missing. These were treated as non-event placeholders, rather than missing measurements for an identified conflict. Every analysis country-year has a corresponding row in the source conflict panel after shifting the year. Country-years with no qualifying conflict were coded 0. No mortality or covariate values were imputed.

Results and checks

The final table contains 3,720 rows and 17 columns, with one row per country-year. There are 691 exposed country-years (using the previous year's conflict), 310 earthquake country-years and 325 drought country-years.

Check	Result
Countries	186
Years	2000–2019
Duplicate country-year keys	0
Missing disaster/conflict indicators	0
Output comparison	All values and missing-value locations agree

The source files used in this assignment differ from the paper's historical analysis sample. The supplied 186-country panel was retained; the paper's 181-country sample and published counts were not imposed.

Comparison of approaches

Both R scripts were run using R 4.3.3, tidyverse 2.0.0 and janitor 2.2.1. After sorting rows and aligning columns, ``all.equal()`` returned ``TRUE``. The outputs agree in all values and missing-value locations. Both approaches therefore produce the same checked analysis table.

The main script explicitly joins the four mortality tables and uses grouped summaries to build the binary indicators. The alternative uses ``map()`` and ``reduce()`` for mortality tables, ``pivot_wider()`` for disasters, and filters qualifying conflicts before extracting unique country-years.

Each script was timed five times in the same R session, with packages already loaded and execution order alternated. Timings include reading source files, processing and writing the output. Median elapsed time was 0.274 seconds for the main script and 0.287 seconds for the alternative. The small difference and overlapping measurements do not establish a general speed advantage. The full measurements are in ``runtime_comparison.csv``.

For these four outcomes, the main script is preferred because the joins and indicator definitions are easy to inspect, and it was slightly faster by median elapsed time in this check. The alternative is easier to extend to additional mortality outcomes and retains fewer conflict rows before joining, but that did not produce a clear overall runtime improvement here. No data differences were found, so no correction was required after comparison.

Files and use

- ``scripts/create_final_data.R``: main preparation script.
- ``scripts/create_final_data_agent.R``: separate alternative script.
- ``scripts/compare_data.R``: checks the two output tables.
- ``data/processed/final_data.csv``: final analysis data.
- ``data/processed/final_data_agent.csv``: comparison output.
- ``reports/comparison_instructions.txt``: task specification for the separate script.

Run the two preparation scripts from the project root with ``tidyverse`` and ``janitor`` installed, then run ``compare_data.R``.